

0 6 . 1 Mutation is one cause of genetic variation in organisms.

Give **two** other causes of genetic variation.

[2 marks]

1 _____

2 _____

In a species of flowering plant, the **T** allele for tallness is dominant to the **t** allele for dwarfness. In the same species, two alleles **C^R** (red) and **C^W** (white) code for the colour of flowers. When homozygous red-flowered plants were crossed with homozygous white-flowered plants, all the offspring had pink flowers.

0 6 . 2 Name the relationship between the two alleles that code for flower colour.

[1 mark]

0 6 . 3

A dwarf, pink-flowered plant was crossed with a heterozygous tall, white-flowered plant.

Complete the genetic diagram to show all the possible genotypes and the ratio of phenotypes expected in the offspring of this cross.

[3 marks]

Phenotypes of parents: Dwarf, pink-flowered × Tall, white-flowered

Genotypes of parents: _____

Genotypes of offspring: _____

Phenotypes of offspring: _____

Ratio of phenotypes: _____

0 6 . 4

A population of this species of plant contained 9% of red-flowered plants.

Use the Hardy–Weinberg equation to calculate the percentage of pink-flowered plants in this population.

Show your working.

[2 marks]

Answer _____ %

8

